# T3 group factors: UV

Active theory: SM + $S_1$ + $S_2$ + $F$. The same beta function is shown across runs in a single comparison table, with exact analytic group factors in the run columns.

$$j = \frac{d-1}{2}, \qquad C_2(d) = \frac{d^2-1}{4}, \qquad T(d) = \frac{d(d^2-1)}{12}.$$

In every comparison table, $\mathcal{S}$ denotes the RGE structure multiplying the displayed group factor in $16\pi^2\beta$.

# Notation key

| symbol | interaction / definition |
|---|---|
| $g_1$ | $D_\mu \supset i g_1 Y B_\mu$ |
| $g_2$ | $D_\mu \supset i g_2 T^A W_\mu^A$ |
| $g_3$ | $D_\mu \supset i g_3 t^A G_\mu^A$ |
| $Y_u$ | $-\bar{Q} Y_u \widetilde{H} u + \text{h.c.}$ |
| $Y_d$ | $-\bar{Q} Y_d H d + \text{h.c.}$ |
| $Y_e$ | $-\bar{L} Y_e H e + \text{h.c.}$ |
| $y_1$ | $L F S_1 + \text{h.c.}$ |
| $y_2$ | $L F S_2 + \text{h.c.}$ |
| $M_F$ | $-\bar{F} M_F F$ |
| $T$ | $\mathrm{Tr}(Y_e^\dagger Y_e + 3 Y_u^\dagger Y_u + 3 Y_d^\dagger Y_d)$ |
| $T_\nu^{(1)}$ | $\mathrm{Tr}(y_1^\dagger y_1)$ |
| $T_\nu^{(2)}$ | $\mathrm{Tr}(y_2^\dagger y_2)$ |
| $T_\nu$ | $T_\nu^{(1)} + T_\nu^{(2)}$ |
| $m_1^2$ | $m_1^2 S_1^\dagger S_1$ |
| $m_2^2$ | $m_2^2 S_2^\dagger S_2$ |
| $\lambda_1$ | $\frac{1}{2}\lambda_1 (H^\dagger H)^2$ |
| $\lambda_{S_1}^{(1)}$ | $\frac{1}{2}\lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$ |
| $\lambda_{S_2}^{(1)}$ | $\frac{1}{2}\lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$ |
| $\lambda_{HS_1}^{(1)}$ | $\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$ |
| $\lambda_{HS_2}^{(1)}$ | $\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$ |
| $\lambda_{HS_1}^{(A)}$ | $\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$ |
| $\lambda_{HS_2}^{(A)}$ | $\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$ |
| $\lambda_5$ | $\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$ |
| $\lambda_{12}^{(1)}$ | $\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$ |
| $\lambda_{12}^{(A)}$ | $\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$ |
| $\lambda_{12}^{(\times)}$ | $\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$ |
| $\lambda_{S_1}^{(A)}$ | $\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$ |
| $\lambda_{S_2}^{(A)}$ | $\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$ |
| $\alpha$ | $Y_{S_1} = \alpha/2,\ Y_F = (\alpha+1)/2,\ Y_{S_2} = (\alpha+2)/2$ |
| $C_2(d)$ | $(d^2-1)/4$ |
| $T(d)$ | $d(d^2-1)/12$ |
| $C_{LLS_1S_2}^{ij;AB}$ | $L_i L_j S_1^A S_2^B$ |
| $C_{ijab}$ | $L_i L_j \phi_a \phi_b$ |
| $C_5^{ij}$ | $(L_i^T C L_j) H H$ |
| $R_W$ | $16\pi^2 \beta_{C_5} \supset R_W \lambda_5 C_{LLS_1S_2}$ |

### Field representations

| run | $S_1$ | $S_2$ | $F$ | $\alpha$ |
|---|---|---|---|---|
| T3 − C | $(2, \frac{1}{2})$ | $(2, \frac{3}{2})$ | $(3, 1)$ | 1 |

## Gauge and Yukawa beta functions

$\beta_{g_1}^{(1)}$

| $\mathcal{S}$ | T3 − C ($\alpha = 1$) |
|---|---|
| $g_1^3$ | $\frac{41}{2}$ |

$\beta_{g_2}^{(1)}$

| $\mathcal{S}$ | T3 − C ($\alpha = 1$) |
|---|---|
| $g_2^3$ | $\frac{31}{6}$ |

$\beta_{g_3}^{(1)}$

| $\mathcal{S}$ | T3 − C ($\alpha = 1$) |
|---|---|
| $g_3^3$ | $-7$ |

$\beta_{Y_u}^{(1)}$

| $\mathcal{S}$ | T3 − C ($\alpha = 1$) |
|---|---|
| $(Y_d(Y_d)^\dagger Y_u)_{ij}$ | $-\frac{3}{2}$ |
| $(Y_u(Y_u)^\dagger Y_u)_{ij}$ | $\frac{3}{2}$ |
| $g_2^2(Y_u)_{ij}$ | $-\frac{9}{4}$ |
| $g_3^2(Y_u)_{ij}$ | $-8$ |
| $g_1^2(Y_u)_{ij}$ | $-\frac{17}{12}$ |
| $T(Y_u)_{ij}$ | 1 |

$\beta_{Y_d}^{(1)}$

| $\mathcal{S}$ | T3 − C ($\alpha = 1$) |
|---|---|
| $(Y_d(Y_d)^\dagger Y_d)_{ij}$ | $\frac{3}{2}$ |
| $(Y_u(Y_u)^\dagger Y_d)_{ij}$ | $-\frac{3}{2}$ |
| $g_2^2(Y_d)_{ij}$ | $-\frac{9}{4}$ |
| $g_3^2(Y_d)_{ij}$ | $-8$ |
| $g_1^2(Y_d)_{ij}$ | $-\frac{5}{12}$ |
| $T(Y_d)_{ij}$ | 1 |

$\beta_{Y_e}^{(1)}$

| $\mathcal{S}$ | T3 − C ($\alpha = 1$) |
|---|---|
| $(y_1(y_1)^\dagger Y_e)_{ij}$ | $\frac{3}{4}$ |
| $(y_2(y_2)^\dagger Y_e)_{ij}$ | $\frac{3}{4}$ |
| $(Y_e(Y_e)^\dagger Y_e)_{ij}$ | $\frac{3}{2}$ |
| $g_2^2(Y_e)_{ij}$ | $-\frac{9}{4}$ |
| $g_1^2(Y_e)_{ij}$ | $-\frac{15}{4}$ |
| $T(Y_e)_{ij}$ | 1 |

$\beta_{y_1}^{(1)}$

| $\mathcal{S}$ | $\mathrm{T3-C}\ (\alpha=1)$ |
|---|---|
| $T_\nu^{(1)}(y_1)_{ij}$ | $\frac{3}{2}$ |
| $(y_2(y_2)^\dagger y_1)_{ij}$ | $\frac{3}{4}$ |
| $(Y_e(Y_e)^\dagger y_1)_{ij}$ | $\frac{1}{2}$ |
| $(y_1(y_1)^\dagger y_1)_{ij}$ | $\frac{5}{4}$ |
| $g_2^2(y_1)_{ij}$ | $-\frac{33}{4}$ |
| $g_1^2(y_1)_{ij}$ | $-\frac{15}{4}$ |

$\beta_{y_2}^{(1)}$

| $\mathcal{S}$ | $\mathrm{T3-C}\ (\alpha=1)$ |
|---|---|
| $T_\nu^{(2)}(y_2)_{ij}$ | $\frac{3}{2}$ |
| $(y_1(y_1)^\dagger y_2)_{ij}$ | $\frac{3}{4}$ |
| $(Y_e(Y_e)^\dagger y_2)_{ij}$ | $\frac{1}{2}$ |
| $(y_2(y_2)^\dagger y_2)_{ij}$ | $\frac{5}{4}$ |
| $g_2^2(y_2)_{ij}$ | $-\frac{33}{4}$ |
| $g_1^2(y_2)_{ij}$ | $-\frac{15}{4}$ |

$\beta_{M_F}$

| $\mathcal{S}$ | $\mathrm{T3-C}\ (\alpha=1)$ |
|---|---|
| $(y_1^T y_1^*)M_F$ | $\frac{1}{2}$ |
| $M_F(y_2^\dagger y_2)$ | $\frac{1}{2}$ |
| $g_2^2 M_F$ | $-12$ |
| $g_1^2 M_F$ | $-6$ |

$\beta_{m_1^2}$

| $\mathcal{S}$ | $\mathrm{T3-C}\ (\alpha=1)$ |
|---|---|
| $m_1^2\, T_\nu^{(1)}$ | $3$ |
| $M_F^\dagger M_F\, T_\nu^{(1)}$ | $-6$ |
| $\lambda_{S_1}^{(1)}\, m_1^2$ | $6$ |
| $g_2^2\, m_1^2$ | $-\frac{9}{2}$ |
| $g_1^2\, m_1^2$ | $-\frac{3}{2}$ |
| $\lambda_{12}^{(1)}\, m_2^2$ | $4$ |

$\beta_{m_2^2}$

| $\mathcal{S}$ | $\mathrm{T3-C}\ (\alpha=1)$ |
|---|---|
| $m_2^2\, T_\nu^{(2)}$ | $3$ |
| $M_F^\dagger M_F\, T_\nu^{(2)}$ | $-6$ |
| $\lambda_{S_2}^{(1)}\, m_2^2$ | $6$ |
| $g_2^2\, m_2^2$ | $-\frac{9}{2}$ |
| $g_1^2\, m_2^2$ | $-\frac{27}{2}$ |
| $\lambda_{12}^{(1)}\, m_1^2$ | $4$ |

$\beta_{\lambda^{(1)}_{HS_1}}$

| $\mathcal{S}$ | $\mathrm{T3-C}\ (\alpha=1)$ |
|---|---|
| $\lambda^{(1)}_{HS_1}\, T^{(1)}_{\nu}$ | 3 |
| $\mathrm{Tr}(y_1^{\dagger} y_1 Y_e^{\dagger} Y_e)$ | $-3$ |
| $\lvert\lambda_5\rvert^2$ | 6 |
| $g_2^2\, \lambda^{(1)}_{HS_1}$ | $-9$ |
| $g_1^2\, \lambda^{(1)}_{HS_1}$ | $-3$ |
| $\lambda_1\, \lambda^{(1)}_{HS_1}$ | 6 |
| $\left(\lambda^{(1)}_{HS_1}\right)^2$ | 4 |
| $\lambda^{(1)}_{HS_1}\, \lambda^{(1)}_{S_1}$ | 6 |
| $\lambda^{(1)}_{12}\, \lambda^{(1)}_{HS_2}$ | 4 |
| $g_2^4$ | $\frac{9}{4}$ |
| $g_1^4$ | $\frac{3}{4}$ |
| $\left(\lambda^{(A)}_{HS_1}\right)^2$ | $\frac{3}{4}$ |
| $\lambda^{(1)}_{HS_1}\, T$ | 2 |

$\beta_{\lambda^{(1)}_{HS_2}}$

| $\mathcal{S}$ | $\mathrm{T3-C}\ (\alpha=1)$ |
|---|---|
| $\lambda^{(1)}_{HS_2}\, T^{(2)}_{\nu}$ | 3 |
| $\mathrm{Tr}(y_2^{\dagger} y_2 Y_e^{\dagger} Y_e)$ | $-3$ |
| $\lvert\lambda_5\rvert^2$ | 6 |
| $g_2^2\, \lambda^{(1)}_{HS_2}$ | $-9$ |
| $g_1^2\, \lambda^{(1)}_{HS_2}$ | $-15$ |
| $\lambda_1\, \lambda^{(1)}_{HS_2}$ | 6 |
| $\left(\lambda^{(1)}_{HS_2}\right)^2$ | 4 |
| $\lambda^{(1)}_{HS_2}\, \lambda^{(1)}_{S_2}$ | 6 |
| $\lambda^{(1)}_{12}\, \lambda^{(1)}_{HS_1}$ | 4 |
| $g_2^4$ | $\frac{9}{4}$ |
| $g_1^4$ | $\frac{27}{4}$ |
| $\left(\lambda^{(A)}_{HS_2}\right)^2$ | $\frac{3}{4}$ |
| $\lambda^{(1)}_{HS_2}\, T$ | 2 |

$\beta_{\lambda^{(1)}_{12}}$

| $\mathcal{S}$ | $\mathrm{T3-C}\ (\alpha=1)$ |
|---|---|
| $\mathrm{Tr}(y_1^{\dagger} y_1 y_2^{\dagger} y_2)$ | $-\frac{9}{2}$ |
| $\lvert\lambda_5\rvert^2$ | 3 |
| $g_2^2\, \lambda^{(1)}_{12}$ | $-9$ |
| $g_1^2\, \lambda^{(1)}_{12}$ | $-15$ |
| $\left(\lambda^{(1)}_{12}\right)^2$ | 4 |
| $\lambda^{(1)}_{12}\, \lambda^{(1)}_{S_1}$ | 6 |
| $\lambda^{(1)}_{12}\, \lambda^{(1)}_{S_2}$ | 6 |
| $\lambda^{(1)}_{HS_1}\, \lambda^{(1)}_{HS_2}$ | 4 |
| $g_2^4$ | $\frac{9}{4}$ |
| $g_1^4$ | $\frac{27}{4}$ |
| $\left(\lambda^{(A)}_{12}\right)^2$ | $\frac{3}{4}$ |

| | |
|---|---|
| $\lambda_{12}^{(1)}\, T_\nu$ | 3 |

$\beta_{\lambda_5}$

| $\mathcal{S}$ | $\mathrm{T3-C}\ (\alpha=1)$ |
|---|---|
| $\lambda_5\, \lambda_1$ | 2 |
| $\lambda_5\, \lambda_{HS_1}^{(1)}$ | 4 |
| $\lambda_5\, \lambda_{HS_2}^{(1)}$ | 4 |
| $\lambda_5\, \lambda_{12}^{(1)}$ | 2 |
| $g_2^2\, \lambda_5$ | $-9$ |
| $g_1^2\, \lambda_5$ | $-9$ |
| $\lambda_5\, \lambda_{HS_1}^{(A)}$ | $-2$ |
| $\lambda_5\, \lambda_{HS_2}^{(A)}$ | 2 |
| $\lambda_5\, \lambda_{12}^{(A)}$ | $-\frac{1}{2}$ |
| $\lambda_5\, T$ | 2 |
| $\lambda_5\, T_\nu$ | $\frac{3}{2}$ |

## Additional scalar-coupling beta functions

$\beta_{\lambda_{HS_1}^{(A)}}$

| $\mathcal{S}$ | $\mathrm{T3-C}\ (\alpha=1)$ |
|---|---|
| $g_1^2 g_2^2$ | 6 |

$\beta_{\lambda_{HS_2}^{(A)}}$

| $\mathcal{S}$ | $\mathrm{T3-C}\ (\alpha=1)$ |
|---|---|
| $g_1^2 g_2^2$ | 18 |

$\beta_{\lambda_{12}^{(A)}}$

| $\mathcal{S}$ | $\mathrm{T3-C}\ (\alpha=1)$ |
|---|---|
| $g_1^2 g_2^2$ | 18 |

## First threshold

The first EFT GroupFactors report below is selected from the actual threshold plan stored in the run. No assumption is made here about whether $F$, $S_1$, $S_2$, or a degenerate set is integrated out first.